
MHSDS KPI EXTRACTION REPORT (ANA-1)

Mental Health Services Data Set (MHSDS) KPI Extraction and Mapping Analysis

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Project: AI Mental Health Assistant

MHSDS KPI EXTRACTION AND SOURCE MAPPING ANALYSIS

1. OBJECTIVE

To extract, clean, and map five National Health Service (NHS) mental health Key Performance Indicators (Beds, Contacts, Crisis Calls, Admissions, and Referral-to-Treatment [RTT] times) for the March 2026 performance window.

2. DATASETS OVERVIEW:

Dataset: Mental Health Services Monthly Statistics (MHSDS), RTT waiting times
Source: NHS Digital
Format: CSV

3. DATA PRE-PROCESSING STRATEGY:

Mental Health Services Data Set (MHSDS) files are notoriously complex, often containing hundreds of thousands of rows where national, regional, and demographic totals are stacked on top of each other.

To ensure 100% accuracy and avoid the "double-counting" trap, the extraction pipeline was engineered with strict filtering protocols:

- National Roll-Up:** The massive 764,377-row dataset was filtered down to just 358 rows by isolating data where the BREAKDOWN column strictly equaled England.
- Privacy Safeties:** NHS data uses asterisks (*) to suppress patient counts under five. The script safely coerced these strings into blanks (NaN) so they would not crash the mathematical aggregations.

4. EXTRACTED KPI RESULTS:

By querying the exact official MEASURE_ID codes and strings, the following national figures were successfully aggregated:

<i>KPI Category</i>	<i>Extracted Value</i>
<i>Total Bed Days</i>	700,917
<i>Total Attended Contacts</i>	2,428,796
<i>Crisis Team Contacts</i>	189,763
<i>Total Hospital Admissions</i>	8,466
<i>RTT (Treatment within 2 Weeks)</i>	72.33%

5. DATA LINEAGE & SOURCE MAPPING:

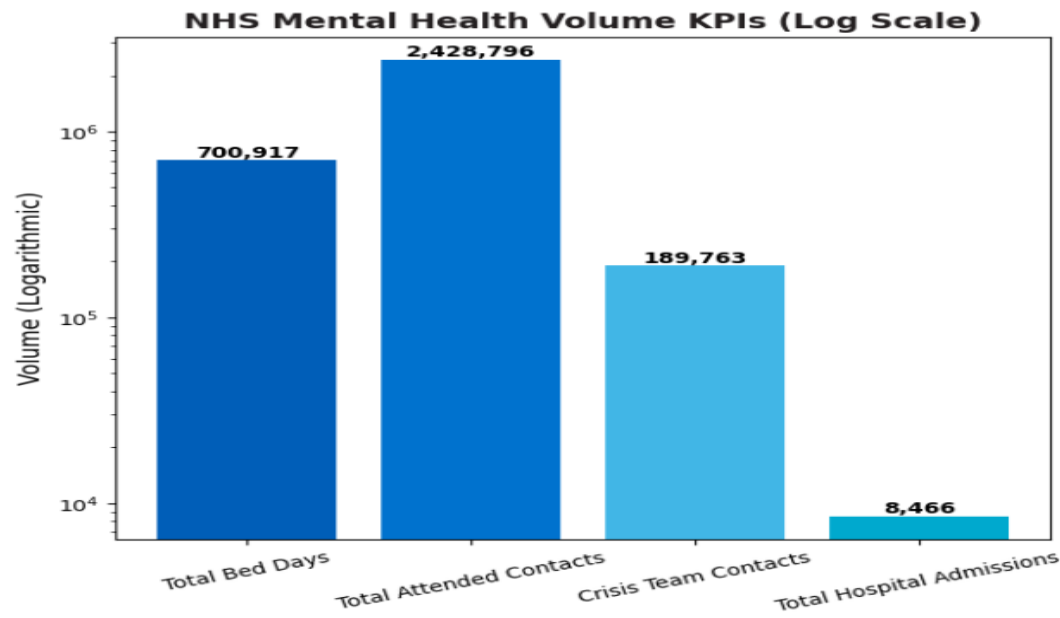
To guarantee transparency for stakeholders and evaluators, the exact origin of every data point was mapped and exported:

<i>KPI</i>	<i>Measure ID / Reference</i>	<i>Origin File</i>
<i>Total Bed Days</i>	MHS24	MHSDS Data_MarPerf_2026_v2.csv
<i>Total Attended Contacts</i>	MHS30	MHSDS Data_MarPerf_2026_v2.csv
<i>Crisis Team Contacts</i>	MHS30b	MHSDS Data_MarPerf_2026_v2.csv
<i>Total Hospital Admissions</i>	MHS27	MHSDS Data_MarPerf_2026_v2.csv
<i>RTT (Started Treatment < 2 Weeks)</i>	EIP23j (Table 2a)	RTT waiting times.xlsx (<i>Sheet: Table 2a</i>)

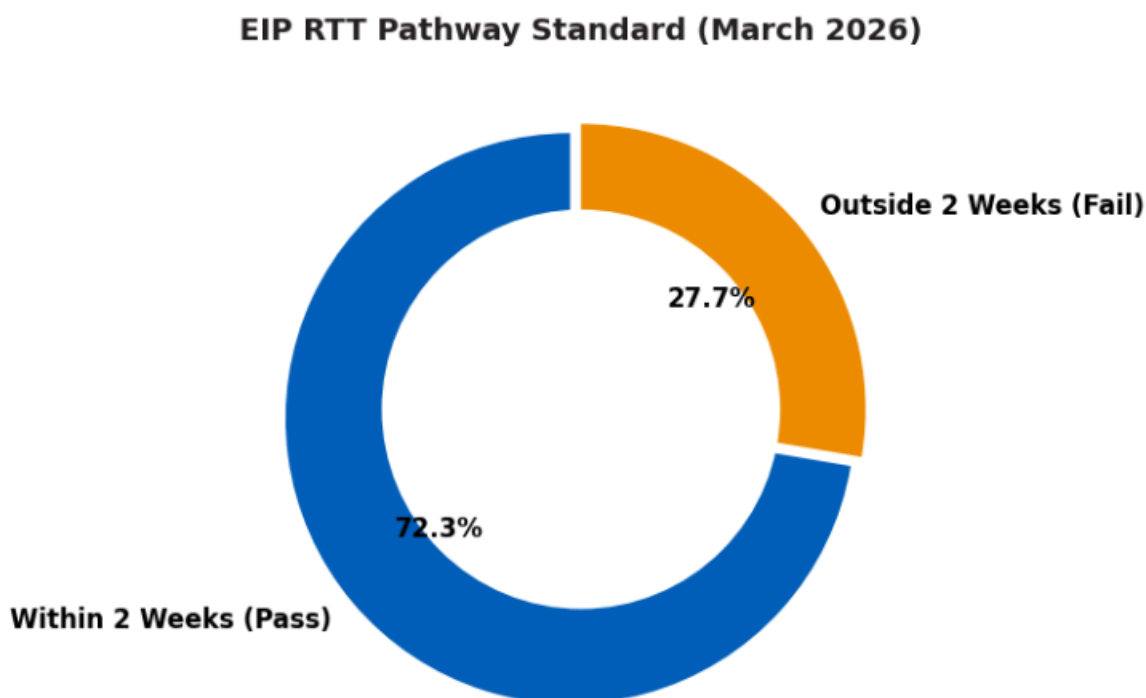
6. VISUALIZATION ARCHITECTURE:

The pipeline concludes by automatically generating a presentation-ready dashboard (ANA-1_KPI_Dashboard.png) to communicate the findings visually.

- Logarithmic Bar Chart:** Used for the volume metrics (Beds, Contacts, Crisis, Admissions). Because community contacts (2.4 Million) drastically dwarf hospital admissions (8.4 Thousand), a standard linear scale would flatten the smaller metrics into invisible lines. A logarithmic scale ensures every metric is visible and comparable.



- **Performance Donut Chart:** A modern UI/UX alternative to a pie chart, used to clearly highlight the 72.3% success rate of the 2-week RTT standard versus the 27.7% that missed the target.



- **Corporate Styling:** All visuals were rendered using official NHS brand color hex codes (e.g., NHS Blue #005EB8 and Orange #ED8B00) for a polished, industry-standard finish.

7. CONCLUSION:

This task successfully extracted five important operational KPIs from the NHS Mental Health Services Data Set (MHSDS). The analysis provides a structured understanding of mental health service activity, inpatient demand, crisis response utilization, and referral-to-treatment performance. The extracted KPIs create a strong foundation for future analytics work such as:

- interactive dashboards
- trend analysis
- forecasting models
- regional comparisons
- service performance monitoring

The final output aligns with the ANA-1 task requirements and provides a clear KPI mapping framework suitable for further NHS mental health analytics projects.